Investigating the Role of CES1 in Non-Cancerous Cells within the Tumour Microenvironment of Colorectal Cancer

Research Project Report

Henry Mileham

Supervised by Daniel D’Andrea

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Abstract

Colorectal Cancer (CRC) is the second leading cause of cancer related mortality worldwide and is driven by complex, heterogeneous, and poorly understood mechanisms. Carboxylesterase 1 (CES1) has previously been identified as a key metabolic mediator in CRC tumour cells that worsens prognosis, particularly in obese patients. The Tumour Microenvironment (TME) is a well established driver of tumourigenesis and immune evasion; however, the role of CES1 in the CRC TME is yet to be explored. This study aimed to identify communication pathways between TME cell types with differential CES1 expression using a Single-Cell RNA Sequencing (scRNAseq) workflow integrated with machine learning algorithms: CellChat and Nichenet. This transcriptomic pipeline revealed the presence of immunosuppressive Ligand-Receptor (L-R) interactions between high-CES1 SPP1 Tumour Associated Macrophages (TAM)s and CD4/CD8 T cells involving the RETN-CAP1, APP-CD74 and CCL20-CCR6 axes. In addition, SPP1 TAMs may promote angiogenesis and tumour growth through communication with high-CES1 Cancer Associated Fibroblast (CAF)s via TGFB and VEGFA stimulation. In contrast, low-CES1 C1QC TAMs are more likely to communicate with Tregs and fibroblasts through tumour-suppressing, antigen-presenting mechanisms driven by MHC-II. In summary, this study demonstrates the key role of CES1 in TME-mediated tumour progression, highlighting the need for experimental validation to establish CES1 as a therapeutic target in CRC.

Keywords: Colorectal Cancer, Tumour heterogeneity, Macrophage, Fibroblast, Tumourigenesis, Immune evasion, Immunotherapy, Single-cell RNA sequencing

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Dedication and Acknowledgements

I would like to thank Daniel D'Andrea for his assistance and expertise that have helped guide me through this project.

Declaration

I declare that the work in this dissertation was carried out in accordance with the requirements of the University's Regulations and Code of Practice for Taught Programmes and that it has not been submitted for any other academic award. Except where indicated by specific reference in the text, this work is my own work. Work done in collaboration with, or with the assistance of others, is indicated as such. I have identified all material in this dissertation which is not my own work through appropriate referencing and acknowledgement. Where I have quoted or otherwise incorporated material which is the work of others, I have included the source in the references. Any views expressed in the dissertation, other than referenced material, are those of the author.

Henry Mileham

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Acronyms

AvEx Average CES1 Expression. 9–11, 27, 30, 33

CAF Cancer Associated Fibroblast. 2, 5, 11–13, 17

CES1 Carboxylesterase 1. 1, 2, 5, 7–19, 25, 27, 30, 32, 33

CRC Colorectal Cancer. 2, 5, 7–9, 12–17, 19, 33

EMT Epithelial Mesenchymal Transition. 14

FAO Fatty Acid Oxidation. 13, 14, 17

FAS Fatty Acid Synthesis. 13, 14

ICC Incoming Communication Celltype. 8, 11, 12, 17, 33

L-R Ligand-Receptor. 2, 5, 7, 8, 10–12, 15–17, 19, 33

N Adjacent-Normal. 7, 9–11, 14, 27, 29

OCC Outgoing Communication Celltype. 8–11, 33

PerEx Percentage of Cells Expressing CES1. 9–11, 27, 30, 33

scRNAseq Single-Cell RNA Sequencing. 2, 7, 9, 13–16, 33

T Tumour. 7, 9–11, 14, 27, 29

TAM Tumour Associated Macrophages. 2, 5, 10–12, 14–17, 19

TME Tumour Microenvironment. 2, 5, 7–10, 12, 14–17, 19

Contents

1 Aims and Objectives

Colorectal Cancer (CRC) is the second leading cause of cancer-related mortality, and a focus of cancer research due to its increasing incidence across a range of demographics [1]. CRC research has been transformed by the introduction of Single-Cell RNA Sequencing (scRNAseq) analysis, which has enabled researchers to characterise intra-tumoural heterogeneity, define cellular transitional states, reconstruct cellular communication networks and quantify gene expression relationships [2–4].

Transcriptomic research done by Capece et al. [5] revealed that expression of the serine hydrolase Carboxylesterase 1 (CES1) in tumour cells is correlated with poorer prognosis in CRC patients. This is attributed to the key regulatory role of CES1 in NF-κB signalling, which has broad impacts on cellular inflammation, metabolic stress adaptation, and oncogenesis. In addition, CES1 has an established metabolic role within myeloid cells [6, 7]. Myeloid cells can be exploited by tumour cells within the Tumour Microenvironment (TME) to drive tumourigenesis and immune evasion [8–11]. Despite this, the mechanisms and pathways by which differential CES1 expression levels in the TME could impact CRC progression remain unexplored. It is plausible that CES1's involvement in lipid metabolism could drive differences in tumourigenesis and immune evasion.

Using an integrated scRNAseq workflow, this study aims to elucidate Ligand-Receptor (L-R) interactions and gene regulatory relationships between cells with differential CES1 expression within the TME. Specific cell-cell interactions were investigated with the hypothesis that high-CES1 cells in the CRC TME could drive tumourigenesis and immune evasion.

1.1 Objective 1: Dataset Selection and Metadata Screening

The primary objective involved preliminary screening and identification of an appropriate CRC scRNAseq dataset with sufficient cell numbers, adequate metadata, and high-quality cells from Tumour (T) and Adjacent-Normal (N) tissue samples. We then screened metadata variables to ensure no bias was introduced during sampling.

1.2 Objective 2: Celltype Characterisation and Identification of Cell Subgroups with Differential CES1 Expression

We aimed to clean, cluster, and analyse CES1 expression across different cell clusters to select the appropriate cell type with high-CES1 expression. Next, we aimed to characterise and select two cell subgroups with differential CES1 expression from within the initially selected cell cluster as the Outgoing Communication Celltype (OCC) for downstream analysis. We aimed to select distinct subclusters with well-established, opposing impacts on tumour progression to create a direct comparison. This would provide insight into mechanisms and pathways that can be altered by these different metabolic states.

1.3 Objective 3: CellChat: Identification of L-R Interactions Between TME Components

Next, we aimed to explore the strength and identity of significant L-R interactions between the OCC, and a communication-receiving cell type within the TME, termed the Incoming Communication Celltype (ICC). This would provide insight into cellular communication patterns that could be associated (positively or negatively) with differential CES1 expression in the OCCs.

1.4 Objective 4: NicheNet: Identification of Gene Regulation Patterns Between Specific Subtypes

Finally, we aimed to explore how gene expression in the ICC was shaped by these priority L-R interactions between subtypes. This would enable detailed insight into the pathways and processes regulated by the interacting ligands from the OCC. Furthermore, this would inform a functional interpretation step undertaken to further understand the context of these relationships in CRC tumour development.

2 Results

2.1 Stromal and Myeloid Clusters Exhibited Elevated CES1 Expression

Initially, using Seurat, the 371,223 high-quality cells from the Pelka et al. [12] CRC scRNAseq dataset were clustered and labelled with the corresponding cell type annotations (Figure 1A). This highlighted seven distinct celltypes: B, Epithelial (Epi), Mast, Myeloid, Plasma, Stromal/Endothelial (Strom), and T/Natural Killer/Innate Lymphoid (TNKILC) cells. Feature plots were used to investigate the distribution of metadata variables: sampling, mismatch repair status, tissue site, sex, tumour stage, and patient identity. Metadata variable analysis revealed negligible bias, including the expected distribution of N and T tissue within clusters, consistent with CRC-scRNAseq literature (Figure 1B). Separation of T and N cells was observed in both the epithelial and stromal clusters, while myeloid cells were predominantly derived from T tissue.

Following screening, Seurat was used to explore CES1 expression within cell type clusters to identify key TME components of interest for communication analysis. Feature plot analysis revealed that CES1-expressing cells resided within specific zones of the Myeloid, Epi, and Strom clusters (Figure 1C). A dot plot displaying the scaled Average CES1 Expression (AvEx) and Percentage of Cells Expressing CES1 (PerEx) within each cluster demonstrated that stromal cells had the highest CES1 expression (AvEx: 2.2, PerEx: 9.5), followed by myeloid (AvEx: 0.2, PerEx: 4.6), then epithelial cells (AvEx: -0.3, PerEx: 3.2) (Figure 1D). Myeloid cells were selected as the OCC for further analysis due to their high CES1 expression, coupled with their established regulatory role in tumour progression and preliminary research implicating CES1-dependent alterations in their function [6, 7, 13].

2.2 Myeloid Group Re-clustering Revealed SPP1/C1QC Macrophage Subgroups

Next, UMAP clustering was carried out on the myeloid cell cluster to identify subgroups within the myeloid population. Cells were labelled according to Pelka et al. [12], who identified ten macrophage subgroups, including M1 (monocyte) and M2 (macrophage-like) classifications (Figure 2A). However, further CES1 analysis required cluster labelling of higher resolution to reveal the novel macrophage subgroups first described by Zhang et al. [14], which included SPP1 and C1QC macrophage groups which have distinct implications for CRC progression. Myeloid cells were clustered into 21 subgroups

(Figure 2B) and analysed using a dot plot following the workflow of Xie et al. [15], which highlighted the expression of markers specific to each myeloid subgroup (Figure 2C).

Clusters expressing high levels of FCN1/S100A8/9 were identified as Macro-FCN1; SPP1/CSTB/SCD2 as Macro-SPP1; C1QC/C1QB/C1QA as Macro-C1QC; and MKI67 as Macro-MKI67. c1/c2/c3/p Dendritic Cells and Cell Doublets were also identified using this method. This enabled the re-labelling of myeloid clusters into 11 distinct subgroups, including C1QC, FCN1, SPP1, and MKI67 macrophages (Figure 2D). These macrophage clusters largely overlapped with the M1/M2 classifications of Pelka et al. [12]; however, the exterior M1/M2 clusters 7 and 17 were reclassified as T and B doublets respectively. The macrophage subgroups could then be used for downstream CES1 analysis on specific subtypes.

2.3 SPP1 TAMs have Significantly Higher CES1 Expression than C1QC TAMs

Next, CES1 expression was explored within each macrophage subgroup to refine the OCC into exemplar macrophage subgroups for cellular communication analysis. AvEx/PerEx dot plots demonstrated that Macro-SPP1 had the highest CES1 expression, followed by Macro-FCN1, with Macro-C1QC showing minimal expression and Macro-MKI67 none at all (Figure 3A). In all macrophage groups, cells expressing CES1 were predominantly derived from T tissue (over 80%) (Figure 3B). Only Macro-C1QC and Macro-SPP1 showed a significant increase in AvEx from cells in N to T tissue, but this increase had a considerably larger effect size for Macro-SPP1 (Figure 3C). Box plots in Figure 3D/E highlight the significant increase in AvEx (median: 0.01 vs 0.07) and PerEx (median: 1.37 vs 7.14) from Macro-C1QC to Macro-SPP1 subgroups, confirming their selection as the refined OCC groups for downstream TME communication analysis.

2.4 RETN-CAP1, APP-CD74, and CCL20-CCR6 L-R Interactions Identified as SPP1-Biased while MHC Class II L-R Interactions Identified as C1QC-Biased

To investigate specific outgoing L-R interactions from SPP1/C1QC macrophages, CellChat circle plots and a bubble plot were utilised. Outgoing SPP1/C1QC macrophage ligands were found to have the highest L-R interaction strength with TNKILC cells (Figure 4A). Of the T cell subgroups, macrophage groups had the strongest interactions with CD8 and Tgd cells (Figure 4B); however, TCD8 and TCD4 were selected for downstream analysis, as preliminary analysis indicated that macrophage-TCD4 interactions were of interest, and further exploratory analysis suggested that Tgd cell interactions were minimal (data not shown). Specific L-R interactions were then mapped, revealing distinct L-R pairs between OCC and ICC types (Figure 4C). The pro-inflammatory host defence : Resistin (RETN), the cytokine: C-C motif chemokine ligand 20 (CCL20), and Amyloid Precursor Protein (APP) ligands all had elevated communication probability in the SPP1 group. MHC class I ligands (HLA-A/B/C/E/F) and Tumour Necrosis Factor (TNFSF12) ligands were expressed equally by both TAMs. MHC class II (HLA-DQ/O/R/M/P) ligands were biased towards C1QC TAMs. Ligands identified using CellChat were used as priority ligands for NicheNet analysis.

2.5 SPP1-Biased L-R Interactions Uniquely Elevate Regulatory Potential in 15 Genes in CD4/CD8 T cells

Priority ligands were then used for NicheNet analysis to identify potential changes in gene regulation between N and T tissues in ICCs (Figure 5A). TNFSF12 dominated gene regulation, activating genes such as NFKB1/2/IA, FAS, CD44, and IL4R across CD4 and CD8 T cell types. The regulatory potential (RP) of C1QC L-R pairs was minimal, while that of SPP1 L-R pairs was moderate, driven mainly by RETN, SPP1, and ICAM1 ligands for genes such as BAX, PTEN, EGR1, and AHSA1. Next, ComplexUpset analysis revealed the distribution of genes distinct to certain cell-cell interactions (Figure 5B). Most genes showed increased regulatory potential when interacting with ligands classified as Both-CD4 and Both-CD8 (47), or Both-CD4 only (26) due to the dominant RP of TNFSF12. Only 15 genes were unique to SPP1 macrophage ligands, including 12 unique to CD4 and 11 unique to CD8. This included elevated RP in genes from RETN (MKNK1, FOXP3, AHSA1, PTEN), APP (UBE2K, PRNP, EGR1, BAX, DDIT3, CCR5, CCL3), ICAM1 (CCL3), SPP1 (JUN, JUND), and CCL20 (CCR6, IFI44) (Figure 5C/D).

2.6 SPP1 TAMs Communicate with High-CES1 CAFs through TGFB and VEGF

Stromal cells not only had high CES1 expression (Figure 1D), but also showed high interaction strength with SPP1 macrophages during initial CellChat analysis (Figure 4A). Therefore, stromal cell subtypes were clustered, assessed for CES1 expression, refined to a single ICC subgroup, and then analysed using CellChat for their L-R interactions with SPP1/C1QC TAMs. Clustering and labelling according to Pelka et al. [12] highlighted the presence of five distinct stromal groups: Endothelial (Endo), Fibroblasts (Fibro), Pericytes (Peri), Schwann cells, and Smooth Muscle cells (Figure 6A). Box plots of AvEx and PerEx showed that the Fibro group had significantly higher CES1 expression than all other stromal subgroups (Figure 6B/C). Alongside endothelial cells, fibroblasts had the highest interaction strength with the two macrophage groups (Figure 6D). The fibroblast subgroup was selected as the ICC due to its significantly elevated CES1 expression, high macrophage-fibroblast interaction strength, and the previously reported interactions between TAMs and CAFs in the CRC TME. L-R pairs were broadly similar to those observed between interacting macrophages and T cells; however, growth factors such as TGFB1, HBEGF, EREG, and VEGFA were also present and showed an SPP1 bias. L-R pairs identified using CellChat were used as priority ligands for NicheNet analysis.

2.7 TGFB and VEGFA Uniquely Elevate Regulatory Potential in 120 Genes

NicheNet analysis of gene RP in fibroblasts highlighted that the RP of C1QC ligands remained minimal, and that "Both" ligands were driven mainly by TNFSF12, similar to the pattern observed in T cells (Figure 7A). However, the SPP1-biased growth factors TGFB1 and VEGFA showed dominant RP across a multitude of genes. ComplexUpset analysis identified 120 genes that were uniquely upregulated by SPP1-biased ligands, highlighting the broad number of unique gene regulations likely triggered by TGFB1 and VEGFA in CAFs (Figure 7B).

3 Discussion

3.1 High CES1 expression in Stromal and Myeloid Cells Could be Driven by Lipid-laden Phenotype

Initial clustering and CES1 analysis shown in Figure 1 demonstrated that CES1 expression was highest in stromal and myeloid populations. Previous scRNAseq analysis of non-cancer tissues suggested that CES1's expression is specific for hepatocytes, monocytes, fibroblasts, and smooth muscle cells [16, 17]. This is consistent with our results other than for smooth muscle cells, which had very low CES1 expression (Figure 6B/C).

In tumour tissues, cell-type-specific differential CES1 expression is yet to be investigated, despite the citation of abnormal CES1 expression in multiple cancers [5, 18–20]. Furthermore, the mechanisms by which CES1 is upregulated in different cell types remain inconclusive. Capece et al. [5] explored the correlation between CES1 levels, adiposity and NF-κB driven inflammation in CRC tumour cells. While causative factors of CES1 upregulation were not studied, they demonstrated the cooperation between a lipid-laden, Fatty Acid Oxidation (FAO) phenotype and CES1 expression, which could be occurring in a similar fashion in myeloid and stromal cells.

Crow et al. [6] highlighted the importance of CES1 in monocyte and macrophage lipid-metabolism. They showed that *in vitro* inhibition of CES1 with oxysterols significantly reduced free cholesterol. In addition, a separate study tied ces1d expression to macrophage differentiation and an inflammatory/metabolic state [7]. Similar to the mechanism proposed by Capece et al. [5], Shao et al. [7] showed that ces1d knockout in obese mice caused an increase in accumulation of triacylglycerol and lipid droplets, but also found this triggered pro-inflammatory activation in macrophages, which enhances migration, activation, and polarisation towards an M1-like phenotype. While these studies do not explain CES1 upregulation in myeloid cells, they do propose the correlation between a lipid-metabolising macrophage state and CES1 expression, which could explain the elevated myeloid CES1 expression seen in our CES1 analysis.

In stromal cells, specifically Cancer Associated Fibroblast (CAF)s, a similar lipid-driven metabolic state has been proposed. Adams et al. [21] explains how in CAFs, a CD36-regulated metabolic switch from a Fatty Acid Synthesis (FAS) state to a FAO state occurs in response to obesity and lipid accumulation. Interestingly, it was proposed that the two lipo-metabolic states could opposingly drive tumour growth through lipid supplementation via FAS or aid angiogenesis and metastasis through FAO. A probe into whether FAO signatures were associated with CES1 expression showed no correlation (data not shown). This null result does not rule out the hypothesis that elevated CES1 expression in stromal cells could be guided by different lipo-metabolic states and warrants a more sensitive follow-up test. Further knockout studies of key proteins in FAO and FAS such as CD36, paired with scRNAseq, could reveal whether CES1 expression is dependent on these opposing lipo-metabolic states in the CRC TME.

3.2 Differential Expression in SPP1/C1QC TAMs

CES1 analysis in macrophage subgroups showed expression levels were significantly higher in SPP1 TAMs compared to C1QC TAMs (Figure 3). SPP1 TAMs, which are enriched in tumour-hypoxic regions, are known to promote Epithelial Mesenchymal Transition (EMT), angiogenesis, and metastatic dissemination [15]. In addition, they are central mediators of intercellular communication in the TME and play a role in coordinating metabolically adapted, therapy-resistant tumours [15, 22–24]. In contrast, C1QC TAMs are context-dependent, mainly driving a tissue-resident, antigen-presenting phenotype through MHC-II expression and immunosenescence [25]. In addition, higher abundance of C1QC macrophages is associated with improved immunotherapy responsiveness through CD4 T cell activation and downstream immune responses [25].

Elevated CES1 expression is expected in SPP1 macrophages. The SPP1 TAM phenotype is described as lipid-laden, "foamy", and PPARγ-driven, an environment also correlated with CES1 expression in myeloid cells [5, 26]. C1QC macrophages are not usually associated with this lipid-laden phenotype, in line with low CES1 expression [25]. One study, however, shows C1Q+ macrophages are associated with higher CES1 expression in obesity-associated lipid metabolism and PPARγ signalling that drives immunosuppression in lung cancer, highlighting the more nuanced, possibly tumour-regulated effects of C1QC macrophages [27]. This could be investigated further using scRNAseq analysis of different lipo-metabolic signatures within SPP1/C1QC macrophage groups.

In addition to cell-type differences, SPP1 and C1QC macrophages also had significant elevation in CES1 expression from N to T tissues (Figure 3C). In the hypoxic TME in T tissue, TAMs are forced to rely on FAO to generate energy, which could drive this increase in CES1 expression [28]. Therefore, this difference could be guided by the same proposed mechanism by which myeloid and stromal cells upregulate lipid metabolism when tumour associated.

3.3 SPP1/C1QC-Biased Ligands and Corresponding Upregulated Genes Associated with Opposing Effects on CRC Progression

CellChat analysis highlighted that SPP1 macrophages communicate with CD4/CD8 T cells through multiple L-R interactions, most notably RETN-CAP1, APP-CD74, and CCL20-CCR6 (CD4 only) (Figure 4). All three L-R axes and many of their associated genes identified from Figure 5 have been associated with worsened prognosis in cancer [29–37]. Pathway analysis reveals that the association between CES1 expression and RETN/APP/CCL20 expression is partially expected and partially novel.

Upregulation of secreted APP is expected in a high-CES1 phenotype. APP is associated with M2/anti-inflammatory pro-tumour macrophage states, which is associated with PPARγ signalling, a process driven by CES1 [5, 26, 38]. In contrast, RETN and CCL20 upregulation is unexpected, with neither having been associated with SPP1 TAMs or the downstream effects of CES1 expression [39, 40]. The demonstrated association of RETN/CCL20 with the high-CES1 SPP1 macrophage phenotype is therefore a novel finding. Future studies should focus on testing this association by selectively inhibiting PPARγ signalling and triacylglycerol hydrolysis in macrophage populations *in vitro*, and using immunohistochemistry (IHC) to confirm the presence of these ligands.

3.3.1 SPP1 TAMs may promote immune evasion via RETN, APP and CCL20 in T cells

The transcriptomic profile generated from CellChat and NicheNet suggests that SPP1 TAM to CD4/CD8 T cell interactions enhance immune evasion within the TME (Figure 8). Our results demonstrate that the immunosuppressive effects could be driven by three mechanisms, which are characterised by Ghorani et al. [41].

3.3.2 Treg Functional Changes

Firstly, overwhelming evidence from our scRNAseq analysis highlights that APP, RETN, and CCL20 would alter Treg cell functioning and reduce Treg activity *in vivo*. RETN expression inhibits CD4 T cell differentiation into Th1 cells, reducing their tumour-suppressing activity [35, 42]. In addition, the RETN-CAP1-upregulated genes FOXP3 and PTEN are known to cooperate to restrain PI3K signalling, drive the immunosuppressive Treg phenotype, and worsen prognosis [43, 44]. The DDIT3/CHOP gene (regulated by APP) directly represses T-bet in CD8 T cells, impairing cytokine production and cytotoxicity [33]. The CCL20-CCR6 axis is also known to drive Treg glycolytic metabolism and drive suppressive activity [45].

The evidence of T cell activity impairment, widespread across all three prioritised L-R interactions, highlights that this could be a pivotal mechanism driving immunosuppression in high-CES1 macrophages. Using tumour-immune co-cultured assays comparing CES1 and CES1*ko* macrophages, fluorescent live cell imaging could be used to measure real-time lysis of tumour destruction, and to analyse the *in vitro* effect of CES1 expression on T cell activity. By pairing this assay with co-cultures with inhibited lytic mechanisms, e.g. FAS/IFNγ signalling, the identities of the driving lytic mechanisms could be tested and correlated with the L-R interactions from this study.

3.3.3 T cell Exhaustion, Apoptosis and PD-1 Expression

Specifically, the APP-CD74 axis and associated genes could contribute towards immune evasion by impairing T-cell functioning through PD-1 expression, promoting exhaustion, and activating apoptosis. APP-CD74 activity in CD4 T cells has been implicated in T cell migration and proliferation, which, when overexpressed, causes exhaustion [46, 47]. BAX upregulation sensitises T cells to intrinsic apoptosis, causing depletion of T cells and immunosuppression [48]. Furthermore, gastric cancer scRNAseq data showed that APP expression in lipid-associated TAMs enhanced macrophage PD-1 expression, which drives immune evasion [49]. The AHSA1 gene, upregulated by RETN, has also been correlated with PD-1 upregulation, causing muted chemotherapy response in head and neck squamous cell carcinoma [50]. PD-1 acts as a brake for immune destruction and is a key focus of research due to its implications for immunotherapy [51]. Should high-CES1 macrophages increase expression of PD-1 *in vivo*, this could have large implications for CRC immunotherapy.

3.3.4 Mediating TME Infiltration

The final mechanism by which the priority ligands may alter immune evasion and immunosuppression, is through regulating the influx of cells into the TME to maximise the immunosuppressive cell content. APP has been shown to restrict TME immune cell influx and impair CD8 T cell immunity by dampening IFN response [52]. APP-upregulated cellular prion protein (PRNP) promotes Treg development and is linked to immunosuppressive cell recruitment and EMT in CRC specifically [29, 53]. The CCL20-CCR6 axis is known to promote therapeutic resistance, metastasis, and immunosuppression by enhancing Treg recruitment and differentiation, while the CCL3-CCR5 axis (upregulated by CCL20, Figure 5) recruits immunosuppressive stromal/Treg populations and is directly linked to CRC progression [36, 54, 55]. Preliminary research also indicates that AHSA1 overexpression is linked to lower CD8 T cell infiltration [30].

In summary, the elevated expression of RETN, APP, and CCL20, and the increased RP of associated genes, indicates the existence of a tumourigenic, immunosuppressive SPP1 macrophage to CD8/CD4 T cell network that could be mediated by elevated CES1 expression (Figure 8).

3.3.5 SPP1 TAMs may initiate angiogenesis and CAF differentiation via TGFB and VEGFA

Figure 6 highlights how interaction strength between macrophages and stromal cells was highest between SPP1 macrophages and fibroblasts. Many of the L-R interactions from SPP1 macrophages were similar to those of T cells, including RETN-CAP1 and SPP1. However, the stark difference lies in the addition of multiple SPP1 TAM-biased growth factors, including TGFB1 and VEGFA, which act on multiple different receptors.

NicheNet analysis showed that regulatory potential was dominated by genes upregulated through TGFB1 and VEGFA. TGFB is the best-established CAF-activating signal in CRC, driving fibroblast and myofibroblast conversion [56–58]. VEGFA is primarily a CAF output, but CRC-specific evidence shows CAF subpopulations actively liberate and respond to micro-environmental VEGFA to drive extracellular matrix deposition and angiogenesis, supporting tumour growth [59].

The presence of SPP1-biased VEGFA and TGFB suggests that high-CES1 macrophages may drive CAF activity, angiogenesis, and tumour growth through the CRC TME. Interestingly, these growth factors are known to be upregulated in the FAO phenotype established by Adams et al. [21], which drives angiogenesis and VEGFA production in CAFs. Despite this, no mechanism connecting CES1 expression to TGFB and VEGFA upregulation has been proposed. Spatial transcriptomics could validate the association between high-CES1 macrophages, high-CES1 fibroblasts, and upregulated angiogenesis and tumour growth. This could test whether a CES1-mediated TAM-CAF pipeline in the TME drives CRC progression.

3.3.6 C1QC TAMs likely contribute towards tumour suppression and antigen presentation

In general, C1QC-biased ligands had a smaller effect on gene RP in T cells and fibroblasts than SPP1-biased ligands. In addition, no C1QC-unique genes were identified in either ICC. However, CellChat analysis highlighted C1QC TAMs' known tendency towards tumour suppression through their well-established upregulation of MHC-class II machinery: HLA-DMB and HLA-DQA4. MHC-class-II antigen-presenting ligands are known to contribute significantly to tumour immunogenicity by increasing CD4 T cell infiltration, delaying T cell exhaustion, and synergising with PD-1 [60]. These effects cooperate to promote tumour regression, increase responsiveness to immunotherapy, and improve patient prognosis [60]. Tumour cells attempt to downregulate this machinery to evade immune detection and weaken the cytotoxic and immune response [60]. Our findings, which show elevated expression of tumour-suppressing machinery within the low-CES1 C1QC macrophages, are not only consistent with the literature, but further demonstrate the likely association between CES1 expression and tumour-promoting effects.

4 Conclusion

In conclusion, this study successfully demonstrated the association between TAM subtypes differentially expressing CES1 and distinct communication patterns in the CRC TME. This work expands on Capece et al. [5]'s finding that elevated CES1 expression in tumour cells drives tumour progression and worsened prognosis. We established that tumour-promoting SPP1 macrophages have elevated CES1 expression, likely relating to their lipid-laden phenotype, which may drive immune evasion through RETN, APP, and CCL20 L-R communication with CD4/CD8 T cells. In addition, our findings suggest that high-CES1 macrophages preferentially communicate with high-CES1 stromal populations through TGFB and VEGFA, promoting tumourigenesis. This is likely opposed by low-CES1 C1QC populations, which preferentially communicate through tumour-suppressing MHC-II mechanisms. Despite these SPP1/C1QC macrophage-biased L-R interactions, the dominant signalling axis between macrophages and T cells identified by NicheNet was Tumour Necrosis Factor signalling, suggesting that T cell activation and apoptosis are the main communication present between these cells [61]. However, this does not negate that minor differences in macrophage-T cell communication can have profound effects on tumour growth and patient prognosis.

Transcriptomic analysis of this kind is vulnerable to dataset-level generalisability issues. Replication in an independent dataset would strengthen generalisability. In addition, these mechanisms were revealed through predictive machine learning algorithms, CellChat and NicheNet, and can only be used to guide future research rather than as direct evidence of interactions. Therefore, the existence of interactions, mechanisms, and *in situ* protein expression must be validated through experimental methods before assumptions can be made about the TME-driven tumour-promoting effects of CES1. Key studies, such as knockout of key ligands, spatial transcriptomics, and IHC analysis, could validate these results. The pairing of experimental methods could identify the underlying biological pathways that dictate CES1 expression and the downstream mechanisms that regulate immune evasion and tumourigenesis.

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5 Figures and Tables

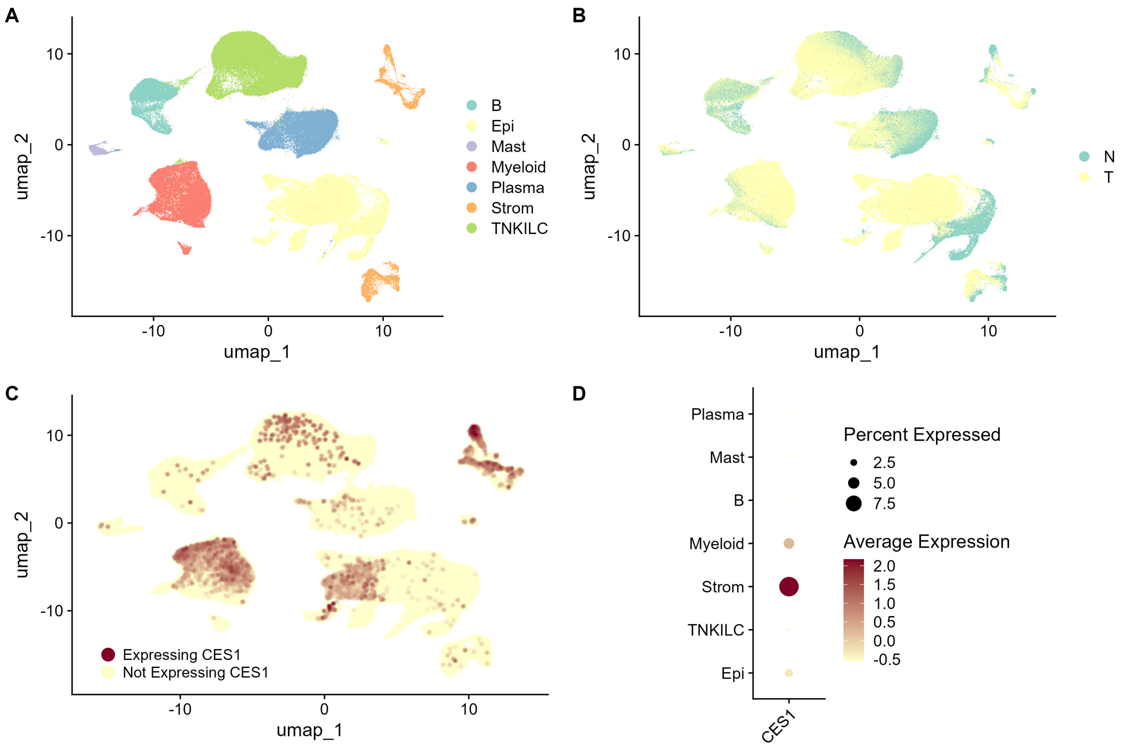

Figure 1: **Elevated CES1 expression present in Stromal and Myeloid populations.** *(A) UMAP plot of 7 cell subtypes with labels from Pelka et al. [12]. (B) Clustering distribution of cells from N and T samples. (C) Clustering distribution of cells expressing CES1. (D) CES1 expression analysis of labelled cell clusters.*

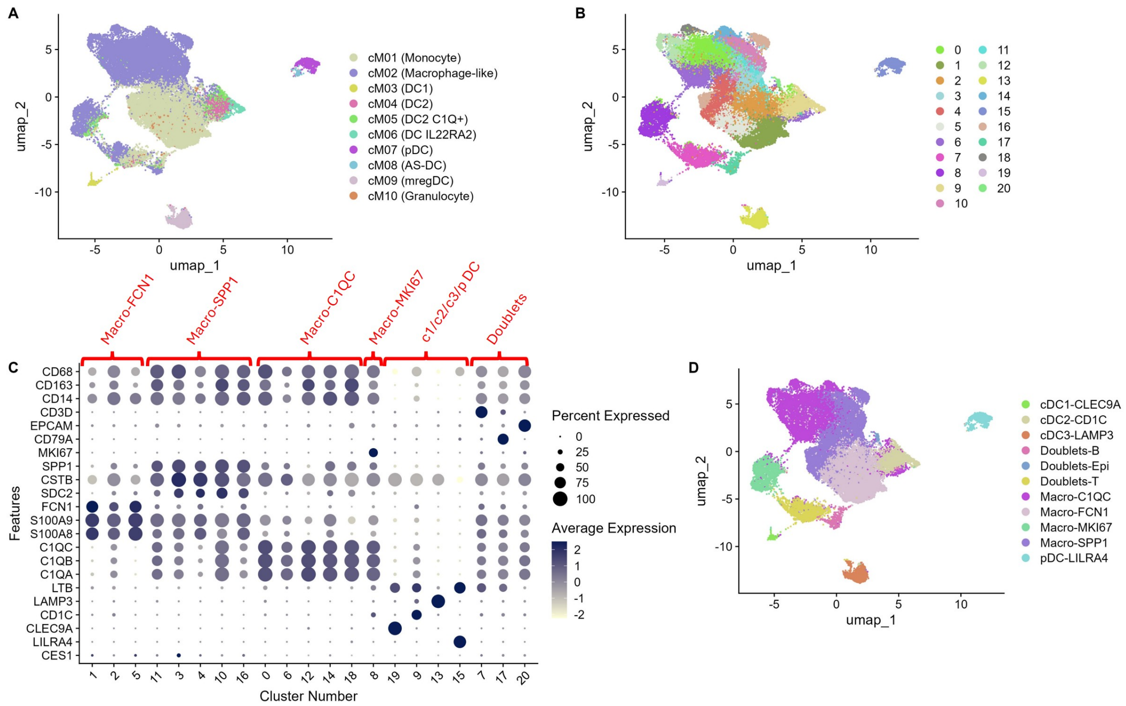

Figure 2: **Identification of Myeloid subsets in CRC.** *UMAP plot labelled with distinct myeloid clusters from (A) Pelka et al. [12], (B) using cluster numbers, and (D) using annotations obtained using feature identification dot plots (C) of the scaled average expression of key myeloid marker genes following Xie et al. [15]'s workflow.*

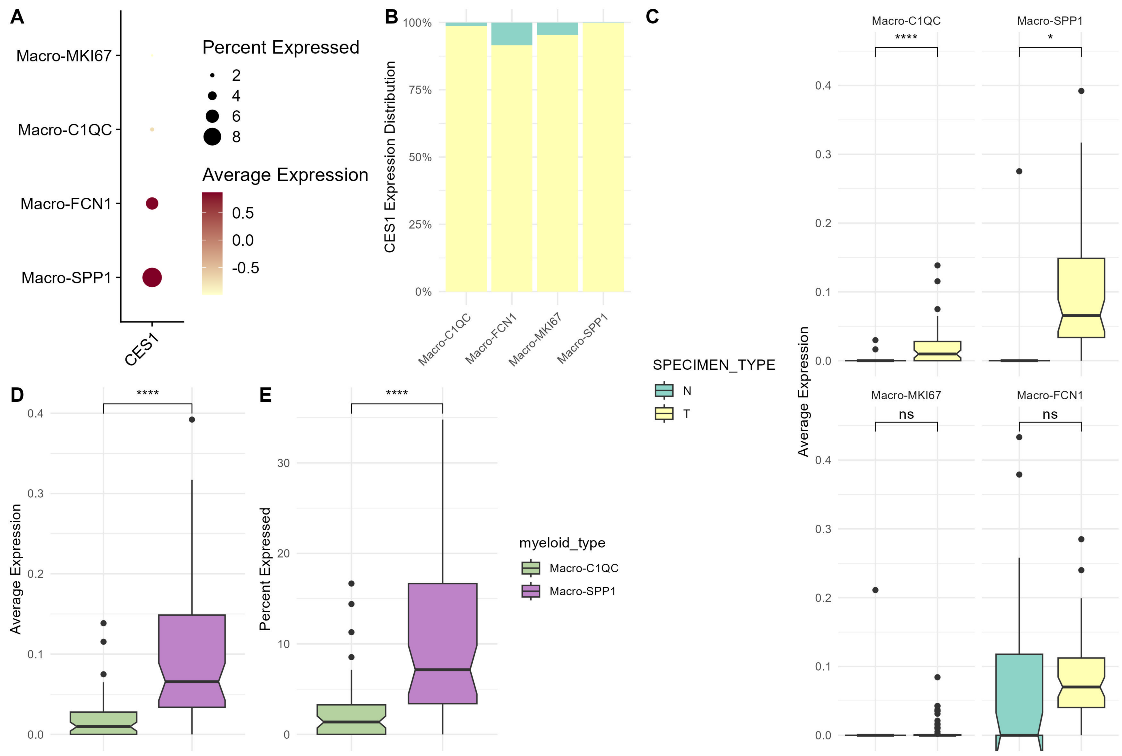

Figure 3: **SPP1 macrophages exhibit significantly higher CES1 expression comparative to C1QC+ macrophages.** *(A) CES1 expression analysis between macrophage groups. (B) Percentage distribution of cells between N and T tissue which express CES1 in macrophage subgroups. (C) Box plots of the comparative AvEx between N and T tissue for macrophage subgroups. (D,E) Comparative AvEx and PerEx between SPP1 and C1QC macrophage groups. ns-Non Significant,*-P<0.05, ****-P<0.0001, Paired T-test.*

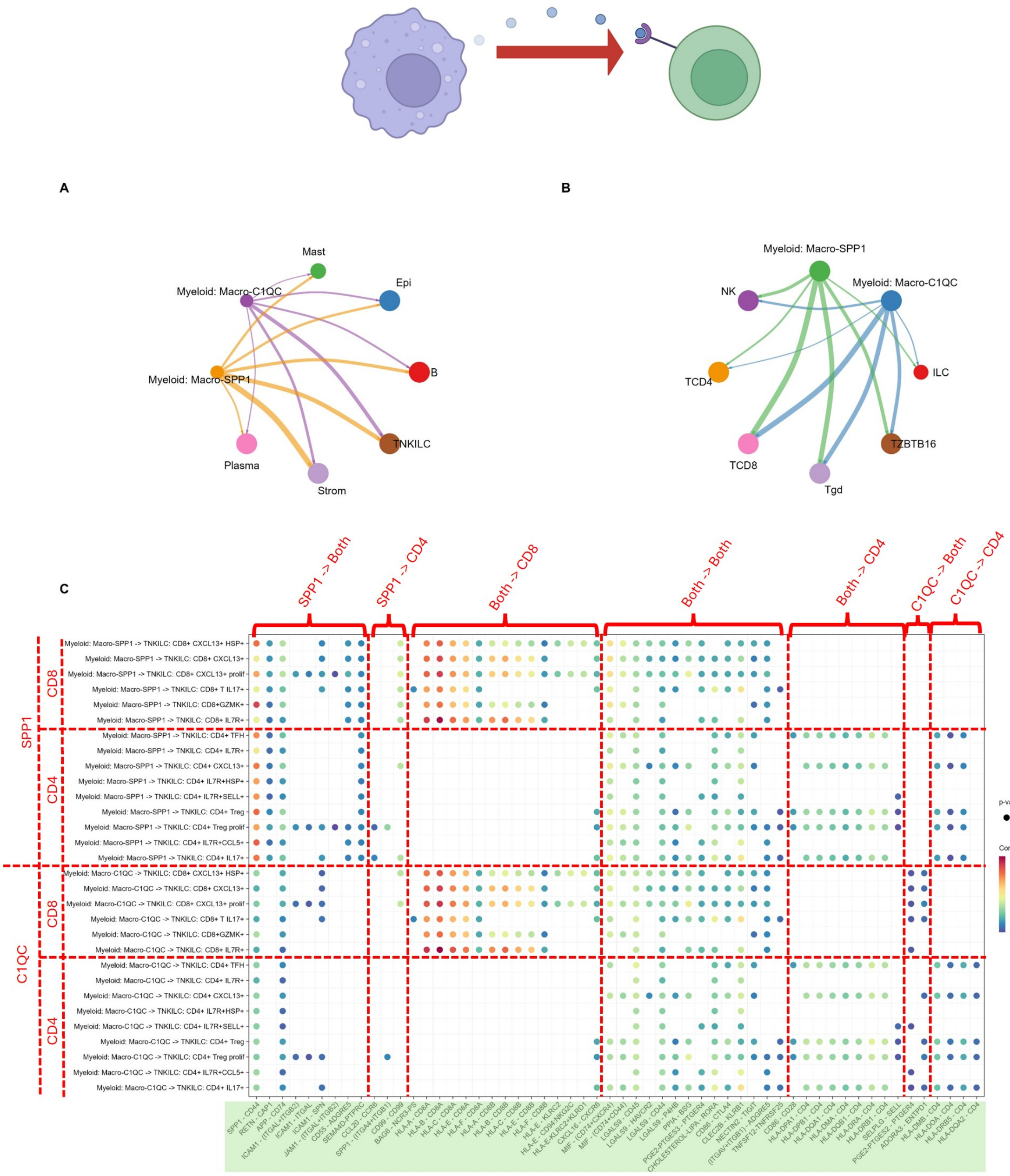

Figure 4: **SPP1/C1QC Macrophage outgoing Ligand-Receptor Interactions.** *CellChat circle plot comparing outgoing interaction strength between SPP1/C1QC macrophages and general cell types (A) and T cell subtypes (B). (C) Pathway level comparison of Ligand-Receptor interaction communication probability (CP) between SPP1/C1QC macrophages and CD4/CD8 T cell subtypes. Interaction labelled "SPP1" when (SPP1-CP)μ > 2(C1QC-CP)μ and vice versa. All combinations in-between classified as "both". This is repeated for CD4/CD8 classifications. Cartoon created using biorender.*

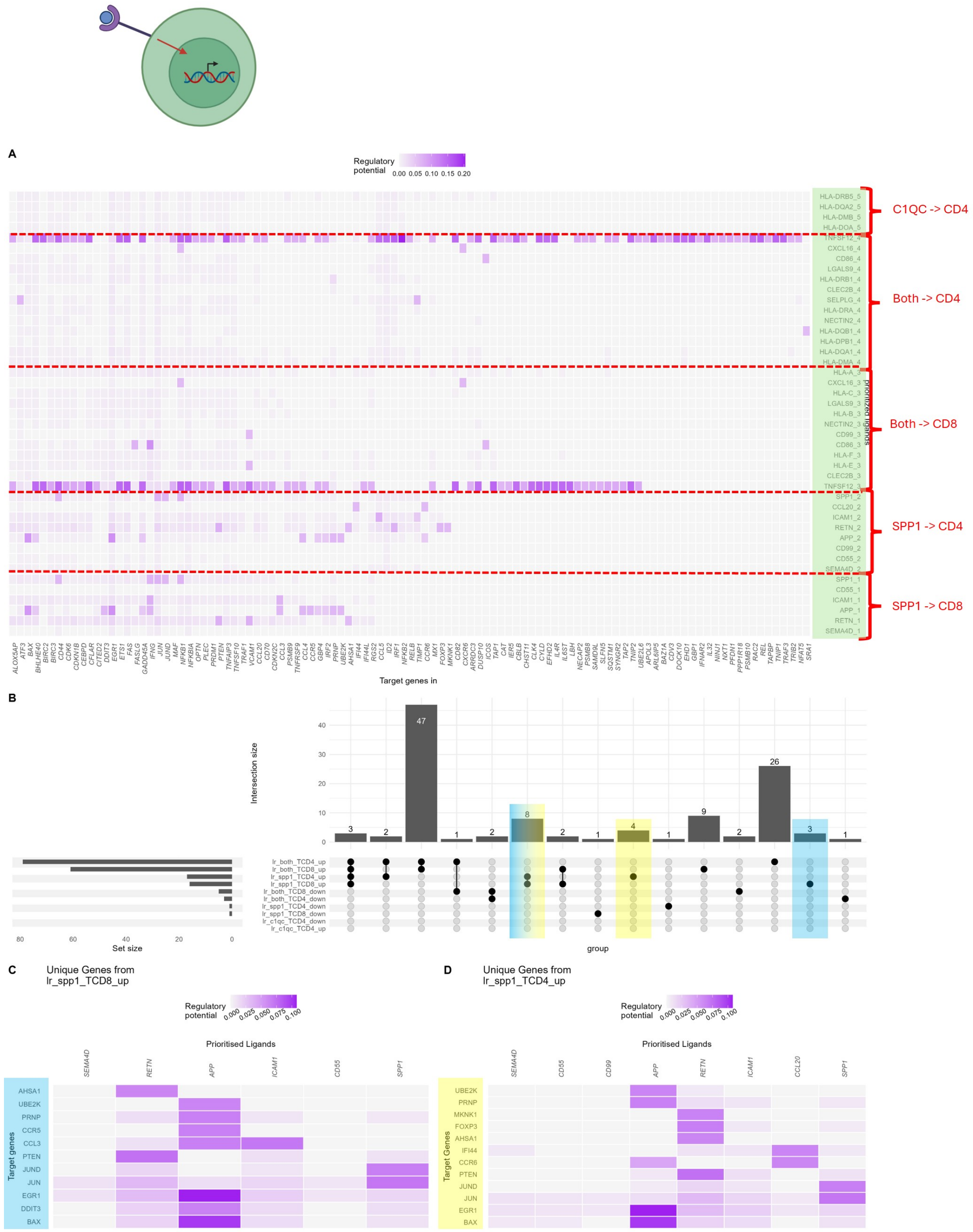

Figure 5: **SPP1 Macrophages could enhance immunosuppressive signalling pathways in T-Cells through RETN, APP and CCL20.** *(A) Heatmap of RP of prioritised ligands between SPP1/C1QC macrophages and CD4/CD8 T cells in N vs T tissue samples. Interactions labelled "SPP1" when (SPP1-CP)µ > 2(C1QC-CP)µ and vice versa. All combinations in-between classified as "Both". (B) ComplexUpset plot showing overlap of genes within each condition. RP Heatmap filtered to include genes uniquely upregulated via specific SPP1-TCD8 (C) or SPP1-TCD4 (D) labelled interactions (no unique gene upregulations present for C1QC-TCD8/TCD4 labelled interactions). Cartoon created using biorender.*

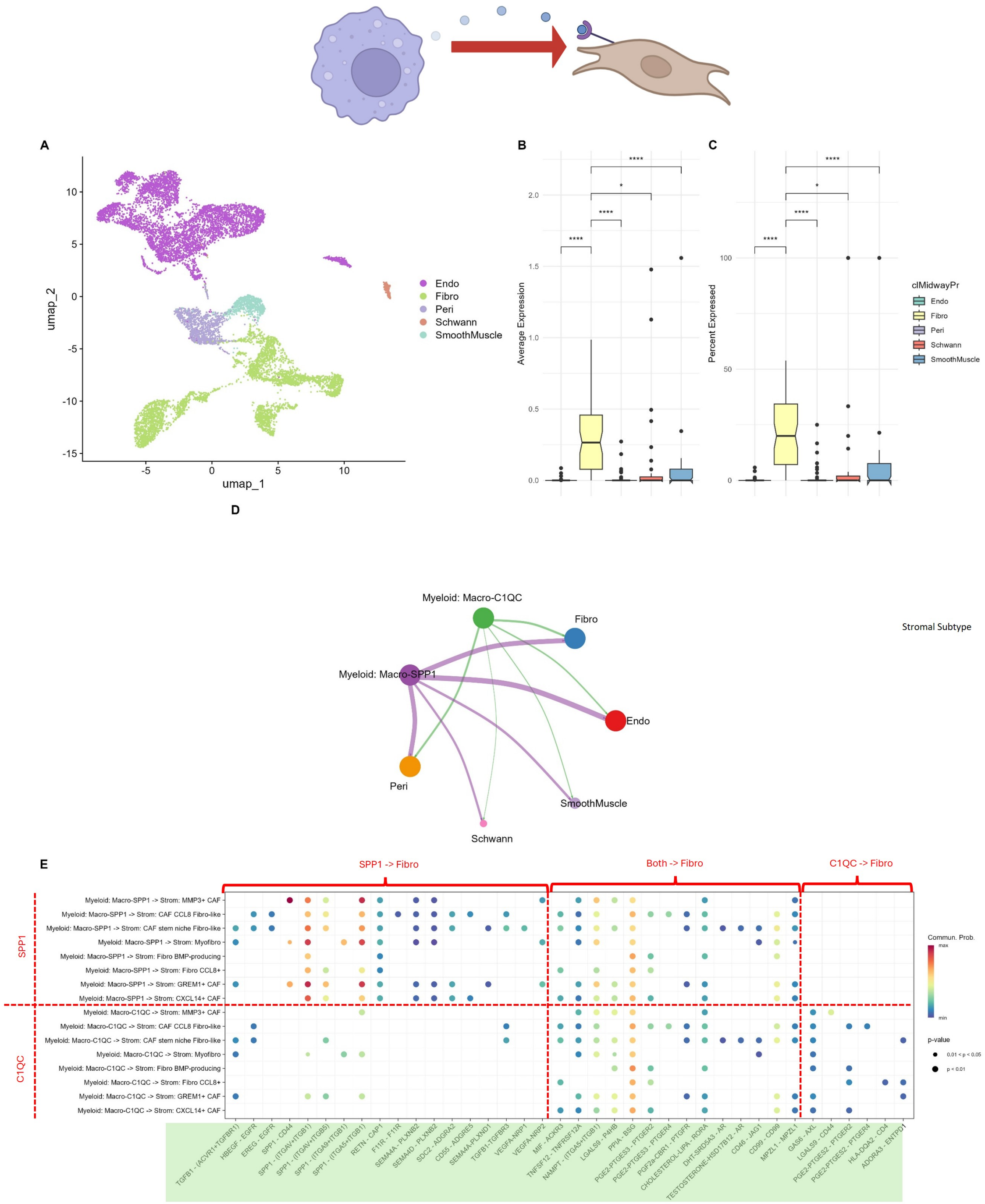

Figure 6: **SPP1/C1QC Macrophage to high-CES1 Fibroblast L-R analysis.** *(A) UMAP plot of 5 distinct stromal clusters with labels from Pelka et al.(B,C) Comparative AvEx and PerEx between stromal subgroups: ns-Non Significant, *-P<0.05,****-P<0.0001, Paired T-test. (D) CellChat circle plot comparing outgoing interaction strength between SPP1/C1QC macrophages and stromal cell subtypes. (E) Pathway level comparison of Ligand-Receptor interaction communication probability (CP) between SPP1/C1QC macrophages and Fibroblast subtypes. Interaction labelled "SPP1" when (SPP1-CP)µ > 2(C1QC-CP)µ and vice versa. All combinations in-between classified as "both". Cartoon created using biorender.*

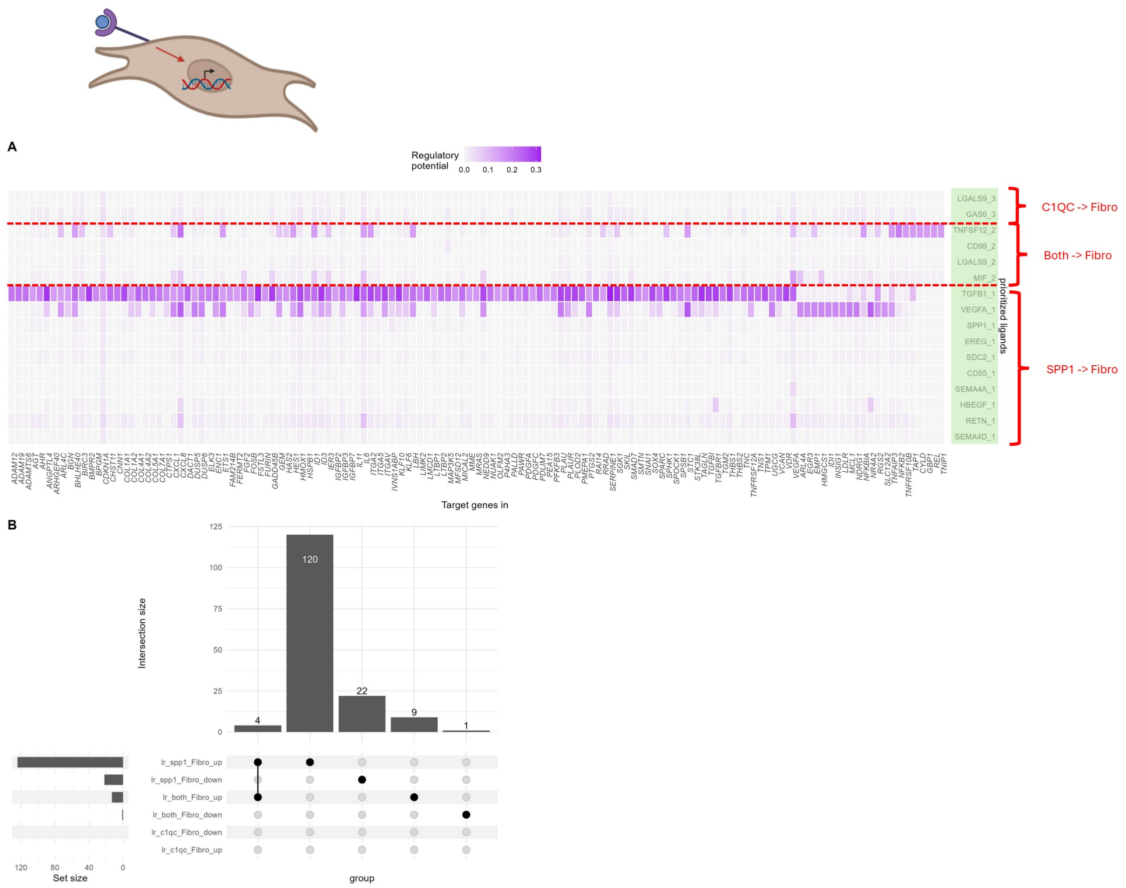

Figure 7: **SPP1 Macrophages upregulate TGFB1 and VEGFA signalling.** *(A) Heatmap of gene regulatory potential (RP) of prioritised ligands between SPP1/C1QC macrophages and Fibroblasts in N vs T samples. Interaction labelled "SPP1" when (SPP1-CP)µ > 2(C1QC-CP)µ and vice versa. All combinations in-between classified as "both". (B) ComplexUpset plot showing overlap of genes within each condition. Cartoon created using biorender.*

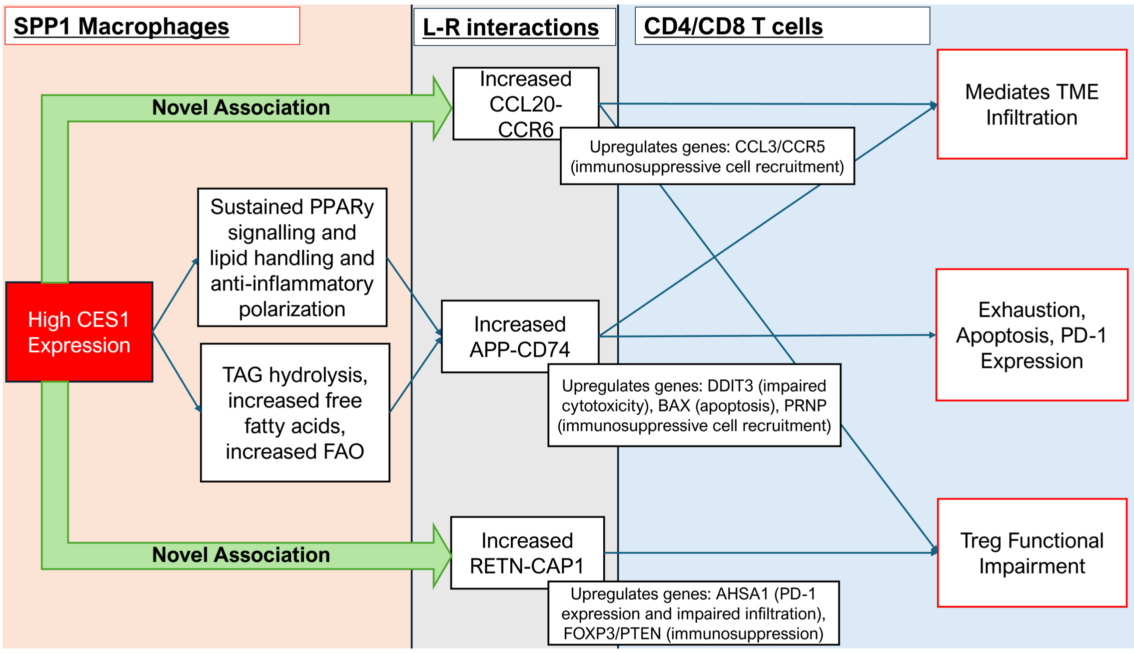

Figure 8: **Proposed mechanism of CES1 mediated tumourigenesis and immune evasion from SPP1 Macrophage – T cell communication.**

6 Supplementary Material

6.1 Methods

6.1.1 Data Collection and Basic scRNAseq Analysis

CRC scRNAseq dataset including metadata and cluster data loaded from from Pelka et al. [12] (GSE178341). Filtering, scaling, normalisation, UMAP clustering and expression difference analysis followed Xie et al. [15]'s workflow using the R package Seurat (version 5.5.0) and all cells were labelled according to Pelka et al. [12] [62]. Xie et al. [15]'s workflow was utilised for stromal and myeloid cell groups. Myeloid cells were re-annotated following Xie et al. [15]'s workflow based on canonical marker genes. CES1 expression scores were aggregated to one value per patient before testing Average CES1 Expression (AvEx) and Percentage of Cells Expressing CES1 (PerEx) to avoid pseudoreplication. Paired t-tests were applied for statistical comparison.

6.1.2 CellChat

CellChat (version 2.2.0.9001) analysis was run from SPP1/C1QC macrophage groups to both CD4/CD8 T cell and fibroblast groups [63]. Custom CellChatDB matrix from Capece et al. [5] was used. Following exploration, ComputeCommunProb(nboot = 100, raw.use = TRUE) was used and visualised using NetVisual_circle() and NetVisual_bubble() including only OCC to ICC interactions. L-R interactions were classified as "SPP1-biased", "C1QC-biased" or "Both" using a 2-fold difference threshold between the average communication probability between cell types to distinguish between L-R interactions biased to certain cell types.

6.1.3 NicheNet

Classifications generated following CellChat were utilised for NicheNet (version 2.2.1.1) analysis [64]. Differential expression was measured using FindMarkers within each ICC population with thresholds: min.pct = 0, p_val_adj $\leq$ 0.05, avg_log2FC $\geq$ 0.25, run separately for up and down regulated genes. Background genes expression $\geq 10\%$ of receiver cells, and "Sender-focused" ligands selected if ligands expressed in $\geq 10\%$ of sender cells with a cognate receptor expressed in $\geq 10\ \%$ of receiver cells and included in default NicheNet human ligand target matrix. Ligand activity predicted using predict_ligand_activities(); ligands ranked by corrected AUPR and all ligands taken forward. get_weighted_ligand_target_links(); 300 gene targets used per ligand and visual threshold utilised top third of regulatory potential weights. Run for all sender/receiver combinations for up and down regulated genes (12 total- T cells, 6 total- Fibroblasts). Ligand-target regulatory potential matrices combined into one matrix and grouped using ComplexHeatmap (version 2.26.1) for genes regulations unique to each cell combination [65].

6.1.4 GitHub

GitHub Repository: CES1–integrative-transcriptomic-analysis